

Connected Peck posets with non-log-concave antichain polynomials

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Abstract

Ding and Dong conjectured that the antichain generating polynomial of every finite connected Peck poset is log-concave. We give an infinite family of height-one counterexamples. For integers $a, c \geq 1$, the poset $P_{a,c}$ has two equally sized ranks and a connected bipartite Hasse graph with a perfect matching. We prove that it is Peck and that its antichain polynomial is

$$\mathcal{N}_{P_{a,c}}(x) = (1 + 2x)^a + 2(1 + x)^{a+c} - 2(1 + x)^a.$$

For every $a \geq 7$, the specialization $P_{a,2}$ violates log-concavity. The first member on this parameter line has 18 elements and log-concavity defect -72 .

1 Construction and statement

Ding and Dong state as Conjecture C that the antichain generating polynomial of every finite connected Peck poset is log-concave. Here “height one” means that maximal chains have one cover edge, or equivalently two rank levels; Ding and Dong number those levels as ranks 1 and 2.

Our construction was motivated by Bhattacharyya and Kahn’s nonunimodal independence-polynomial construction for a bipartite graph. Their two bipartition classes have unequal sizes, so that graph does not directly give a rank-symmetric Peck poset. The balanced rank layers, the added perfect matching, and the closed formula below are the features used here to obtain a connected Peck counterexample family.

Let $a, c \geq 1$. Define a two-rank poset $P_{a,c}$ with lower and upper ranks

$$L = A \sqcup W, \quad R = B \sqcup C,$$

where

$$|A| = |C| = a, \quad |W| = |B| = c.$$

The cover relations are all pairs in $A \times B$, the matching pairs $A_i < C_i$, and all pairs in $W \times (B \sqcup C)$.

Theorem 1. *For all $a, c \geq 1$, the poset $P_{a,c}$ is a finite connected Peck poset and*

$$\mathcal{N}_{P_{a,c}}(x) = (1 + 2x)^a + 2(1 + x)^{a+c} - 2(1 + x)^a. \quad (1)$$

For every $a \geq 7$, the coefficient sequence of $\mathcal{N}_{P_{a,2}}(x)$ is not log-concave. Consequently, the connected-Peck log-concavity conjecture is false.

2 The Peck property

The Hasse graph is connected because $a, c \geq 1$. Both ranks have size $a + c$, so the poset is rank-symmetric and rank-unimodal. The edges

$$\{A_i C_i : 1 \leq i \leq a\} \cup \{W_j B_j : 1 \leq j \leq c\}$$

form a perfect matching. By König's theorem, the maximum independent-set size of the bipartite Hasse graph is $a + c$. Antichains are exactly independent sets of this graph, while either rank is an antichain of size $a + c$; hence $P_{a,c}$ is Sperner.

Every element has a neighbour in the other rank, and no chain contains three elements. Thus all maximal chains have two elements. For $k \geq 2$, the whole $2(a + c)$ -element poset is a maximum k -family and its size equals the sum of the two largest rank sizes. Therefore $P_{a,c}$ is strongly Sperner. It is consequently Peck under the definition used by Ding and Dong.

3 The antichain polynomial

We count independent sets according to whether they meet W . If no vertex of W is chosen, the remaining graph consists of the complete bipartite graph between A and B , together with the matching from A to C . Its independence polynomial is

$$(1 + 2x)^a + (1 + x)^{a+c} - (1 + x)^a.$$

Indeed, the first term counts choices that use no vertex of B , while the second counts choices that use no vertex of A ; their overlap consists of arbitrary subsets of C and is exactly $(1 + x)^a$, which explains the subtraction.

If a nonempty subset of W is chosen, no vertex of $B \sqcup C$ may be chosen, while the vertices of A remain arbitrary. This contributes

$$((1 + x)^c - 1)(1 + x)^a = (1 + x)^{a+c} - (1 + x)^a.$$

Their sum is (1).

4 An infinite family of failures

Set $c = 2$, and write $p_i = [x^i]\mathcal{N}_{P_{a,2}}(x)$. The top three coefficients are

$$p_{a+2} = 2, \quad p_{a+1} = 2(a + 2), \quad p_a = 2^a - 2 + (a + 2)(a + 1).$$

Log-concavity at position $a + 1$ would require

$$2^{a+1} \leq 2(a+2)(a+3) + 4. \quad (2)$$

Define

$$F(a) = 2^{a+1} - 2(a+2)(a+3) - 4.$$

Then $F(7) = 72 > 0$, and

$$F(a+1) - F(a) = 2^{a+1} - 4(a+3) > 0$$

for every $a \geq 7$. Thus (2) fails for all $a \geq 7$.

At $a = 7$, the polynomial is

$$1 + 18x + 114x^2 + 378x^3 + 742x^4 + 882x^5 + 602x^6 + 198x^7 + 18x^8 + 2x^9,$$

and the final internal log-concavity defect is

$$18^2 - 198 \cdot 2 = -72.$$

Direct enumeration of all 2^{18} vertex subsets agrees coefficientwise with (1). The same verification for $a = 1, \dots, 6$ shows that $a = 7$ is the first failure on the $c = 2$ parameter line. We do not claim global minimality among all connected Peck posets.

References

- [1] A. Bhattacharyya and J. Kahn, *A bipartite graph with non-unimodal independent set sequence*, Electronic Journal of Combinatorics **20** (2013), Paper 11. [journal page](#).
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